Algorithm 1 LLM multi-agent workflow for constructing the Citywalk mobility-theme-affect network

Input: Citywalk notes $X = \{x_i\}_{i=1}^{n}$, LLM model `qwen-plus`, LLM temperature $\tau = 0$, Amap API $\gamma(\cdot)$, POI aggregation threshold $r = 100\,\text{m}$, and maximum distance retained between adjacent POIs $d_{\max} = 1{,}500\,\text{m}$

Output: Directed, weighted geographical network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, w_e)$

Set LLM $\leftarrow$ `qwen-plus`; instantiate $\mathcal{A}^{\text{LLM}}_{\text{spatial}}$, $\mathcal{A}^{\text{LLM}}_{\text{theme}}$, and $\mathcal{A}^{\text{LLM}}_{\text{PAD}}$ with specialized roles and structured output schemas; initialize $\mathcal{S} \leftarrow \emptyset$

for all Citywalk notes $x_i \in X$ **do**

$P_i \leftarrow \mathcal{A}^{\text{LLM}}_{\text{spatial}}(x_i)$ ▷ *Agent 1: spatial trajectory coding*

$P_i = \{(\text{location_index}_{ij}, \text{location_POI}_{ij}, \text{text_evidence}_{ij})\}_{j=1}^{m_i}$ ▷ *structured, chronological array of location points*

$t_i \leftarrow \mathcal{A}^{\text{LLM}}_{\text{theme}}(x_i, P_i)$ ▷ *Agent 2: walking theme categorization*

$t_i = (\text{Primary_Theme}_i, \text{Theme_Reasoning}_i)$ ▷ *one Primary_Theme representing the dominant focus*

$A_i \leftarrow \mathcal{A}^{\text{LLM}}_{\text{PAD}}(x_i, P_i)$ ▷ *Agent 3: PAD-based affective appraisal*

$A_i = \{(\text{P_score}_{ij}, \text{A_score}_{ij}, \text{D_score}_{ij}, \text{is_inherited}_{ij}, \text{reasoning}_{ij})\}_{j=1}^{m_i}$ ▷ *affective appraisal for each featured POI*

$P_i \leftarrow \text{SemanticFilter}(P_i)$ ▷ *screen POI names for geocodability*

$Z_i \leftarrow \gamma(P_i, \text{“Xiamen”})$ ▷ *geocoding and coordinate transformation*

$S_i \leftarrow \text{Join}(P_i, Z_i, t_i, A_i)$ ▷ *link through shared note and location identifiers*

Append S_i to $\mathcal{S}$

end for

$\mathcal{V} = \{v_{ij} = c_r(s_{ij}) : s_{ij} \in \mathcal{S}\}, \qquad \bar{\mathbf{a}}_v = \dfrac{1}{|v|} \displaystyle\sum_{s_{ij} \in v} \mathbf{a}_{ij}$ ▷ *consolidate POIs and average their PAD vectors*

$\mathcal{E} = \{(v_{ij}, v_{i,j+1}) : v_{ij} \neq v_{i,j+1},\ d(v_{ij}, v_{i,j+1}) \leq d_{\max}\}$ ▷ *link consecutively reported POIs after spatial aggregation*

$w_e(u, v) = \displaystyle\sum_{i=1}^{n} \sum_{j=1}^{|S_i|-1} \mathbb{I}[(v_{ij}, v_{i,j+1}) = (u, v)]$ ▷ *cumulative count of linkages between a specific pair of nodes*

$\mathcal{G} \leftarrow (\mathcal{V}, \mathcal{E}, w_e)$

return $\mathcal{G}$